

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5- Pseudocode Design	Assessment:	Assessment Tool 1 – Pseudocode Design
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	V.Naga Nischay	Reg. No.:	1924722
Section / Batch:	CSE(AI)/Slot D	Date:	25-08-2026

Code of Conduct

I P.Poorna Chandra(192472287) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Poorna Chandra

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution	20	Innovative,	Logical	Basic solutions	Incorrect or no	
Design		feasible solutions with strong justification	solutions with adequate justification	with weak justification	solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Q11. Linear Regression – Training, Prediction, Residuals, MSE and R²

BEGIN

Input dataset with features X and target values Y

Split dataset into training data and testing data

Initialize linear regression model

Train the model using training data

Calculate predicted values:

$Y_{\text{pred}} = \text{Model.predict}(X_{\text{test}})$

Calculate residuals:

$\text{Residual} = Y_{\text{test}} - Y_{\text{pred}}$

Calculate Mean Squared Error:

$\text{MSE} = \text{Average}((Y_{\text{test}} - Y_{\text{pred}})^2)$

Calculate R² Score:

$$R^2 = 1 - \frac{\text{Sum}((Y_{\text{test}} - Y_{\text{pred}})^2)}{\text{Sum}((Y_{\text{test}} - \text{Mean}(Y_{\text{test}}))^2)}$$

Display predicted values

Display residuals

Display MSE

Display R² score

END

Q12. Logistic Regression – Pass/Fail Prediction

BEGIN

Input student dataset containing:

Study Hours

Attendance Percentage

Result (Pass/Fail)

Convert Pass = 1 and Fail = 0

Select Study Hours and Attendance as input features X

Select Result as target Y

Split the dataset into training and testing data

Initialize Logistic Regression model

Train the model using training data

Input study hours and attendance of a new student

Predict the probability of passing

If probability ≥ 0.5

Predict "PASS"

Else

Predict "FAIL"

Display prediction

END

Q13. Logistic Regression – Sigmoid Probability and Binary Class
BEGIN

Input feature values X

Calculate linear value:

$$Z = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Calculate sigmoid probability:

$$P = 1 / (1 + e^{(-Z)})$$

Set threshold = 0.5

If $P \geq 0.5$

Class = 1

Else

Class = 0

Display sigmoid probability P

Display binary class

END

Example:

If probability = 0.78

$0.78 \geq 0.5$

Class = 1

If probability = 0.32

$0.32 < 0.5$

Class = 0

Q14. Logistic Regression – Evaluation Metrics

BEGIN

Input actual labels Y_{test}

Predict labels using Logistic Regression model

Create confusion matrix

Calculate:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Calculate Accuracy:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Calculate Precision:

$$\text{Precision} = TP / (TP + FP)$$

Calculate Recall:

$$\text{Recall} = TP / (TP + FN)$$

Calculate F1-Score:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Display:

Confusion Matrix

Accuracy

Precision

Recall

F1-Score

END

Q15. Unsupervised Learning – Clustering

BEGIN

Input an unlabelled dataset

Load the dataset

Select required features

Handle missing values

Scale/normalize the features

Initialize clustering algorithm

Example: K-Means

Choose the number of clusters K

Train the clustering model

Discover groups based on similarity

Assign each data point to a cluster

Store cluster assignments

Display the clusters and cluster assignments

END

Simpleflow:

Unlabelled Data → Preprocessing → Feature Scaling → Clustering → Discover Groups → Cluster Assignments